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Burst valve

Abstract

A valve assembly is disclosed that has a valve housing including a first axial end, a cap, a second axial end, and an airflow pathway extending from the first axial end towards the second axial end. The cap is positioned towards the second axial end. The valve housing includes an environmental opening configured for fluid communication with an outside environment, an enclosure opening towards the first axial end configured for fluid communication with an interior of an enclosure, and a film having an expansion diameter disposed in the valve housing laterally across the airflow pathway. The cap has an inner lateral surface and a piercing protrusion (having a piercing tip) extending from the inner lateral surface towards the first axial end. An axial distance is defined between the film and the inner lateral surface. A ratio of the axial distance to the film expansion diameter is at least 0.14.

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Background/Summary

(1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/404,290, filed Sep. 7, 2022, the disclosure of which is incorporated herein by reference.

TECHNOLOGICAL FIELD

(1) The present disclosure is generally related to a valve assembly. More particularly, the present disclosure is related to a burst valve for an enclosure.

BACKGROUND

(2) Various situations require conditions that allow the ability to release pressure from an enclosure. For example, applications within electronics enclosures, such as batteries, that contain sensitive electronic components and equipment, as well as a heat source, often require that pressure levels within the enclosure be maintained and regulated to operate consistently. An improper level of pressure can interfere with mechanical and electrical operations of the components and equipment.

(3) It can also be desirable to prevent contaminants or pollutants from the ambient environment within the enclosed environment. Contaminants can reduce the efficiency and longevity of the components within the enclosure. Such contaminants can enter the enclosure from external sources, and the contaminants can gradually damage the valve or the enclosure, resulting in deterioration of performance and even complete failure of the device. Consequently, the valve assembly may have one or more films capable of removing moisture and contaminants in any air entering the device, such as breathable membranes.

SUMMARY

(4) The technology disclosed herein relates to a valve assembly that is configured to have improved bursting of a film at or above a set pressure differential between the ambient environment and the

enclosure. In some embodiments the technology disclosed herein improves the ability to consistently burst a film at or above a set pressure differential. Improved bursting of film may reduce damage to the enclosure. Such improvements may be realized by having a cap height of the valve assembly such that the film can flex or expand into the cap until the film bursts open. Such improvements may be realized by having a piercing protrusion pierce the film when it flexes or expands. Such improvements may be realized by having a ratio of a film expansion diameter to a distance between the film and an inner surface of the cap such that the film can flex or expand into the distance until the film bursts open.

(5) In one or more embodiments, the valve assembly includes a valve housing. The valve housing includes a first axial end, a cap, a second axial end, and an airflow pathway. The airflow pathway extends from the first axial end towards the second axial end. The cap is positioned towards the second axial end. The valve housing further includes an environmental opening. The environmental opening is configured for fluid communication with an outside environment. The valve housing further includes an enclosure opening towards the first axial end. The enclosure opening is configured for fluid communication with an interior of an enclosure. The valve housing further includes a film. The film is disposed in the valve housing laterally across the airflow pathway. The film is disposed in the valve housing along the airflow pathway between the enclosure opening and the environmental opening. The cap has an inner lateral surface and a piercing protrusion. The piercing protrusion extends from the inner lateral surface of the cap towards the first axial end. The piercing protrusion includes a piercing tip. An axial distance between the film and the inner lateral surface of the cap is greater than 4.4 mm.

(6) In some such embodiments, the axial distance between the film and the inner lateral surface of the cap is at least 7.4 mm. Additionally or alternatively, the piercing protrusion is configured to pierce the film when a pressure differential between the enclosure opening and the environmental opening rises above a burst threshold. Additionally or alternatively, the burst threshold is at least 50 mbar. Additionally or alternatively, a distance between the film and the piercing protrusion is 1 mm to 3 mm. Additionally or alternatively, a length of the piercing protrusion is at least 2 mm.

(7) Additionally or alternatively, the cap has a cap sidewall extending from the inner lateral surface towards the first axial end. The cap sidewall defines a cap height. Additionally or alternatively, the cap has a cap base. The cap sidewall extends from the inner lateral surface to the cap base. The piercing protrusion extends beyond the cap base towards the first axial end by at least 1 mm. Additionally or alternatively, the cap height is at least 2 mm. Additionally or alternatively, the valve housing further includes a housing sidewall extending between the inner lateral surface and the first axial end.

(8) Additionally or alternatively, the valve assembly further includes a void space. The void space is defined among the piercing protrusion, the housing sidewall, the inner lateral surface, and the film. Additionally or alternatively, the film includes a breathable membrane. Additionally or alternatively, airflow through the airflow pathway is configured to increase by at least 12 times after the film is pierced compared to airflow through the airflow pathway before the film is pierced. Additionally or alternatively, the film includes a non-breathable membrane. Additionally or alternatively, a surface area of an expansion area of the film is 450 square mm to 1500 square mm. Additionally or alternatively, the piercing protrusion tapers towards the first axial end.

(9) In one or more embodiments, the valve assembly includes a valve housing. The valve housing includes a first axial end, a cap, a second axial end, and an airflow pathway. The airflow pathway extends from the first axial end towards the second axial end. The cap is positioned towards the second axial end. The valve housing further includes an environmental opening. The environmental opening is configured for fluid communication with an outside environment. The valve housing further includes an enclosure opening towards the first axial end. The enclosure opening is configured for fluid communication with an interior of an enclosure. The valve housing further includes a film. The film is disposed in the valve housing laterally across the airflow pathway. The

film is disposed in the valve housing along the airflow pathway between the enclosure opening and the environmental opening. The film has an expansion diameter. The cap has an inner lateral surface and a piercing protrusion. The piercing protrusion extends from the inner lateral surface of the cap towards the first axial end. The piercing protrusion has a piercing tip. An axial distance is defined between the film and the inner lateral surface. A ratio of the axial distance to the film expansion diameter is greater than 0.135.

(10) In some such embodiments, the axial distance between the film and the inner lateral surface of the cap is at least 7.4 mm. Additionally or alternatively, the ratio of the axial distance to the expansion diameter is at least 0.228. Additionally or alternatively, the piercing protrusion is configured to pierce the film when a pressure differential between the enclosure opening and the environmental opening rises above a burst threshold. Additionally or alternatively, the burst threshold is at least 50 mbar. Additionally or alternatively, a distance between the film and the piercing protrusion is 1 mm to 3 mm. Additionally or alternatively, a length of the piercing protrusion is at least 2 mm.

(11) Additionally or alternatively, the cap includes a cap sidewall extending from the inner lateral surface towards the first axial end. The cap sidewall defines a cap height. Additionally or alternatively, the cap includes a cap base. The cap sidewall extends from the inner lateral surface to the cap base. The piercing protrusion extends beyond the cap base towards the first axial end by at least 1 mm. Additionally or alternatively, the cap height is at least 2 mm. Additionally or alternatively, the valve housing further includes a housing sidewall extending between the inner lateral surface and the first axial end.

(12) Additionally or alternatively, the valve assembly further includes a void space. The void space is defined among the piercing protrusion, the housing sidewall, the inner lateral surface, and the film. Additionally or alternatively, the film includes a breathable membrane. Additionally or alternatively, airflow through the airflow pathway is configured to increase by at least 12 times after the film is pierced compared to airflow through the airflow pathway before the film is pierced. Additionally or alternatively, the film includes a non-breathable membrane. Additionally or alternatively, the film expansion diameter is at least 32.5 mm. Additionally or alternatively, a surface area of an expansion area of the film is 450 square mm to 1500 square mm. Additionally or alternatively, the piercing protrusion tapers towards the first axial end.

(13) The above summary is not intended to describe each embodiment or every implementation. Rather, a more complete understanding of illustrative embodiments will become apparent and appreciated by reference to the following Detailed Description of Exemplary Embodiments and claims in view of the accompanying figures of the drawing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a first perspective view of an example valve assembly consistent with various embodiments.

(2) FIG. 2 is a second perspective view consistent with the example of FIG. 1.

(3) FIG. 3 is a perspective cross-sectional view consistent with the example of FIG. 1.

(4) FIG. 4 is a cross-sectional view consistent with the example of FIG. 1.

(5) FIG. 5 is a schematic facing view of a variety of shapes of expansion areas of films consistent with the technology disclosed herein.

(6) FIG. 6 is a schematic side view of another embodiment of a valve assembly and illustrates increasing pressure.

(7) FIG. 7 is a schematic side view consistent with the example of FIG. 1 and illustrates increasing pressure.

(8) FIG. 8 is an exemplary graph depicting performance data.

(9) FIG. 9 is another exemplary graph depicting performance data.

(10) FIG. 10 is another example valve assembly consistent with various embodiments.

(11) The present technology may be more completely understood and appreciated in consideration of the following detailed description of various embodiments in connection with the accompanying drawings.

(12) The figures are rendered primarily for clarity and, as a result, are not necessarily drawn to scale. Moreover, various structure/components, including but not limited to fasteners, electrical components (wiring, cables, etc.), and the like, may be shown diagrammatically or removed from some or all of the views to better illustrate aspects of the depicted embodiments, or where inclusion of such structure/components is not necessary to an understanding of the various exemplary embodiments described herein. The lack of illustration/description of such structure/components in a particular figure is, however, not to be interpreted as limiting the scope of the various embodiments in any way.

DETAILED DESCRIPTION

(13) Valve assemblies consistent with the technology disclosed herein can have a variety of different configurations. FIGS. 1-4 depict one example embodiment of a valve assembly 100, and FIGS. 1-4 can be viewed together with the following description. A valve assembly 100 is disclosed that generally has a valve housing 16, a film 20, and a piercing protrusion 12.

(14) The valve assembly 100 is configured to be disposed in a receptacle of an enclosure, E, a portion of an example of which is illustrated in FIGS. 3-4. The valve assembly 100 is generally configured to relieve pressure in the enclosure E that is above a threshold pressure differential between the enclosure and the environment outside of the enclosure. In some implementations, the valve assembly 100 can be configured to have improved reliability for pressure release above the threshold pressure differential compared to some existing pressure relief devices. The valve assembly 100 is generally configured to burst open the film 20 such that the film 20 is opened to allow pressure relief. The valve assembly 100 is generally configured to puncture and burst the film 20 when the pressure differential between an enclosure opening 19 and an environmental opening 18 rises above the threshold pressure differential, which is referred to herein as the burst threshold.

(15) The valve housing 16 has a first axial end 1, a cap 10, a second axial end 2 opposite the first axial end 1, an airflow pathway A extending from the first axial end 1 towards the second axial end 2, the environmental opening 18 and the enclosure opening 19. The valve housing 16 is generally configured to contain various components of the valve assembly, including the film 20. The valve housing 16 can be constructed of a variety of different materials and combinations of materials. In some embodiments the valve housing 16 is a molded plastic. In another embodiment the valve housing 16 is a metal. In one example, at least a portion of the valve housing 16 is an injection-molded plastic, 3D printed plastic or other material, etc.

(16) The airflow pathway A is generally defined by the valve housing 16. The airflow pathway A is configured to fluidly connect the environmental opening 18 and the enclosure opening 19. The airflow pathway A can be defined by the valve housing 16 such that the valve housing 16 surrounds the airflow pathway A. The airflow pathway A can also be defined by the cap 10 of the valve housing 16. In some embodiments, the airflow pathway A may allow for airflow in two directions (as illustrated in FIGS. 1, 3, and 4). In alternative embodiments, the airflow pathway A may obstruct airflow from the enclosure opening 19 towards the environmental opening 18 until the film 20 is burst. Once the film 20 is burst, or once the burst threshold is reached, then the airflow pathway A may allow for two-way airflow.

(17) The enclosure opening 19 is generally configured for fluid communication with an interior of the enclosure E. The enclosure opening 19 is generally configured to define a portion of the airflow pathway A. The enclosure opening 19 is positioned towards the first axial end 1. In the current example, the enclosure opening 19 faces the axial direction, meaning that the enclosure opening 19

is perpendicular to a central axis of the valve housing **16**. In alternative embodiments, the enclosure opening **19** can face a lateral direction. The enclosure opening **19** can be defined by a body **16a** of the valve housing **16**. The valve housing **16** can be configured to be sealably coupled to the enclosure E. In some embodiments, the enclosure opening **19** is the only source of airflow A into or out of the enclosure E.

(18) The enclosure opening **19** can define a variety of shapes and sizes. Further, the enclosure opening **19** can include one or more openings that cumulatively define the enclosure opening.

(19) The environmental opening **18** is generally configured for fluid communication with the outside environment, which can be the ambient environment. The environmental opening **18** is generally configured to define a portion of the airflow pathway A. The environmental opening **18** (**18a-c** as illustrated in FIG. 1) can be between the first axial end **1** and the second axial end **2**. In alternative embodiments, the environmental opening(s) **18** can be positioned towards the second axial end **2** relative to the enclosure opening **19**. The environmental opening **18** can include one or more openings. The environmental opening **18** can include, for example, four environmental openings (**18a-c** as illustrated, and an additional opening can be opposite opening **18b**, although it is not visible in FIG. 1), or more than four environmental openings **18**.

(20) The environmental opening(s) **18** can be defined laterally around the valve housing **16**. In alternative embodiments, the environmental opening(s) **18** can be defined by, for example, the cap **10**. The environmental opening **18** can face in the lateral direction, as illustrated in FIGS. 1-4. In alternative embodiments, the environmental opening **18** can face in an axial direction (not illustrated). For example, the environmental opening **18** can be defined at the second axial end **2** and face in the axial direction.

(21) The valve housing **16** is generally configured to engage with the enclosure E (such as, e.g., a battery enclosure) via a mounting structure **38** as illustrated in FIG. 1. Such engagement may be, for example, a bayonet engagement, a threadable engagement, a snap fit engagement, a fastener engagement, or a welded engagement, to sealably connect the valve assembly **100** to the enclosure E. As illustrated in FIGS. 1-4, the valve housing **16** can include a bayonet connector **40** (**40a-d** as illustrated in FIG. 2). The bayonet connector **40** is configured to sealably couple the valve assembly **100** to the enclosure E. The bayonet connector **40** can be configured to engage a mating structure defined by the enclosure E. The bayonet connector **40** can include one or more tabs which extend away from the first axial end **1** of the valve housing **16**. The bayonet connector **40** can define the enclosure opening(s) **19**.

(22) The valve assembly **100** can further include a seal **50** (as illustrated in FIGS. 3-4). The seal **50** can be, for example, an o-ring or gasket. The seal **50** can be rubber or another elastomeric material. In some embodiments, the seal **50** is configured to seal the valve assembly **100**/valve housing **16** to the enclosure E. The seal **50** is generally configured to prevent airflow from bypassing the valve assembly **100**. In alternative embodiments, the seal **50** may be a weld, for example, or an adhesive. The seal **50** can be disposed within a seal receptacle **52** defined by the valve housing **16** as illustrated in FIG. 4. The seal **50** can fully surround the enclosure opening **19**.

(23) The valve assembly **100** can further include the film **20** disposed in the valve housing **16** (as illustrated in FIGS. 3-4). The film **20** is generally configured to burst open and allow pressure relief at or above the burst threshold. Prior to bursting, the film **20** is generally configured to obstruct contaminants from moving into or out of the enclosure opening **19**.

(24) The film **20** is disposed laterally across the airflow pathway A. The film **20** can be disposed along the airflow pathway between the enclosure opening **19** and the environmental opening **18**. The film **20** is generally configured to be coupled to the valve housing **16**. The film **20** can have a perimetric region **20b** that is bonded to the valve housing **16**. The valve housing **16** can define a mounting surface **22** around the airflow pathway A that is sealably coupled to the perimetric region **20b** of the film **20**. The mounting surface **22** can be positioned between the first end **1** and the second end **2**. In some embodiments the perimetric region **20b** is welded to the valve housing **16**. In

other embodiments, the film **20** has a support ring (which may or may not be circular) to support the perimetric region **20b**, and the support ring is coupled to the vent housing. In still other embodiments, the perimetric region **20b** is coupled to the valve housing **16** with adhesive. An expansion area **20a** (partially visible in FIG. 3) of the film **20** is an unbonded area of the film **20** surrounded by the perimetric region **20b**. The expansion area **20a** is expandable in response to a pressure differential between the enclosure opening **19** and the environmental opening(s) **18**. The expansion area **20a** can be characterized by an expansion diameter **21** (illustrated in FIG. 3). The expansion diameter **21** is defined as a diameter across a circle that is circumscribed by the expansion area **20a** of the film **20**.

(25) The expansion diameter **21** defines a region of maximum tension of the film during expansion, which limits the deflection of the film. In the current example, the expansion area **20a** is circular in shape, and the expansion diameter **21** is the diameter of the expansion area **20a**. In one or more alternative embodiments, the expansion area **20a** is not circular, some examples of which are depicted in FIG. 5. In such embodiments, the expansion diameter **21** may be less than a cross-dimension of the expansion area **20a**. In FIG. 5, the perimetric region **20b** is not currently depicted, but would be located around the perimeter of each expansion area **20a**. As discussed above, the perimetric region is configured to be coupled to the valve housing around the airflow pathway.

(26) The surface area of the expansion area **20a** is measured when the expansion area is in an undeflected state, such as when the pressure differential across the film is zero. The surface area of the expansion area can be 450 square millimeters (mm) to 1500 square mm. In alternative embodiments, the surface area of the film **20** can be at least 350 square mm, at least 400 square mm, at least 450 square mm, at least 500 square mm, at least 600 square mm, at least 700 square mm, at least 800 square mm, at least 900 square mm, at least 1000 square mm, at least 1100 square mm, at least 1200 square mm, at least 1300 square mm, at least 1400 square mm, or at least 1500 square mm. In further alternative embodiments, the area of the film **20** can be less than 1500 square mm, less than 1400 square mm, less than 1300 square mm, less than 1200 square mm, less than 1100 square mm, less than 1000 square mm, less than 900 square mm, less than 800 square mm, less than 700 square mm, less than 600 square mm, less than 500 square mm, less than 450 square mm, less than 400 square mm, or less than 350 square mm.

(27) In one or more embodiments, the film **20** can be a breathable membrane. In such embodiments, the film **20** is configured to prevent particles from passing through the enclosure opening **19**. In some embodiments, the film **20** is also configured to prevent liquids passing through the enclosure opening **19**. The film **20** may be configured to filter particulate or debris from the air as it flows along the airflow pathway A. The film **20** may be configured to control humidity in the enclosure E by filtering water from the air as it flows along the airflow pathway A. In some embodiments the film **20** allows passive venting between the enclosure opening **19** and the environmental opening **18** along the airflow pathway A.

(28) In alternative embodiments, the film **20** can be a non-breathable membrane meaning that there is no measurable airflow through the rupture valve **280** at room temperature with 20 mbar pressure differential. In some embodiments the film **20** has a moisture vapor transmission rate of 0.0 at 73° F. and 50% relative humidity in accordance with ASTM E96-00 (2021). In such alternative embodiments, airflow is obstructed between the environmental opening and the enclosure opening **19** along the airflow pathway A. In some embodiments the film **20** defines selective communication between the enclosure opening **19** and the environmental opening **18** along the airflow pathway. Such selective communication can be, for example, pressure relief from the enclosure opening **19** to the environmental opening **18** when the pressure differential reaches or exceeds the burst threshold.

(29) The film **20** can be constructed of a variety of different materials and combinations of materials. In various embodiments the film **20** incorporates a breathable membrane, such as polytetrafluoroethylene (PTFE) or other types of breathable membranes. The film **20** can be a

laminate or composite that includes a breathable membrane, such as a PTFE laminated to a woven or non-woven support layer. In some embodiments, the film **20** is a woven fabric or a non-woven fabric. The film **20** can be constructed of hydrophobic material, or the film **20** can be treated to exhibit hydrophobic properties. In one example, the film **20** is a hydrophobic woven or non-woven fabric. The film **20** can be constructed of an oleophobic material, or the film **20** can be treated to exhibit oleophobic properties. In one example, the film **20** is an oleophobic woven or non-woven fabric. In some embodiments the film **20** is constructed of a non-breathable PTFE. In some such embodiments the film **20** can be an unexpanded PTFE. In some such embodiments the film **20** can be a non-porous PTFE. In some embodiments the PTFE is skived PTFE. In some embodiments the PTFE is a cast PTFE. PTFE materials may advantageously have relatively high thermal and mechanical stability compared to some other materials. Other types of polymeric materials are also contemplated. In various embodiments the film **20** is liquid impermeable.

(30) The valve housing **16** may further include a support brace **60** (as visible in FIGS. 2-3). The support brace **60** may be configured to provide stability and support to the film **20**. In some embodiments the support brace **60** is configured to limit deflection of the film **20** towards the first end **1**. In the embodiment shown, the support brace **60** may include one or more support braces **60** arranged laterally across the airflow pathway A. In alternative embodiments, the support brace(s) **60** can be arranged circumferentially around the airflow pathway A. The support brace(s) **60** can define the enclosure opening(s) **19**.

(31) The valve assembly **100** may include a housing sidewall **14** (illustrated in FIGS. 3 and 4). The housing sidewall **14** is an axially-extending structure that laterally surrounds the airflow pathway A. The housing sidewall **14** generally extends in the axial direction between the first end **1** and the second end **2**. In this example, the housing sidewall **14** extends in the axial direction from the film **20** (or mounting surface **22**) towards the second end **2**. The housing sidewall **14** can be disposed around the central axis X of the valve housing **16**. The housing sidewall **14** is disposed around the film **20**. The housing sidewall **14** is positioned laterally outward from the expansion area **20a**, in some embodiments. The housing sidewall **14** can be configured to partially isolate the airflow pathway A from the surrounding environment. The housing sidewall **14** can be configured to protect the film **20** from direct impact by debris in the outside environment. The environmental opening(s) **18** can be defined by the housing sidewall **14**, laterally around the valve housing **16**.

(32) As mentioned above, the valve housing **16** has the cap **10**. The cap **10** may be configured to protect the interior of the valve assembly **100**. The cap may be configured to protect the film **20** from being directly impacted by debris from the outside environment. The cap **10** can define the second axial end **2** of the valve assembly **100**. In alternative embodiments, the cap **10** can be positioned towards the second axial end **2**, but does not define the second axial end **2**. The cap **10** generally has an inner lateral surface **11** (visible in FIG. 4). The inner lateral surface **11** is generally a relatively planar lateral surface. The inner lateral surface **11** helps define the airflow pathway A.

(33) The valve housing **16** generally has the cap **10** and a valve body **16a** that is coupled to the cap **10**. In the current example, the cap **10** and the valve body **16a** cumulatively define the valve housing **16**. In some embodiments consistent with the example depicted, the valve body **16a** includes a first portion of the housing sidewall **14a** and the cap **10** defines a second portion of the housing sidewall **14b** (see FIG. 4). In such embodiments, the cap **10** is configured to be coupled to the first portion of the sidewall **14a** of the valve body **16a** via the second portion of the sidewall **14b**. In some other embodiments, the cap **10** does not include a portion of the housing sidewall **14**, and the housing sidewall **14** is entirely a component of the valve body **16a**. In yet other embodiments, the housing sidewall **14** is entirely a component of the cap **10** and the valve body **16a** does not include a portion of the housing sidewall **14**. The housing sidewall **14** generally extends between the inner lateral surface **11** and the first end **1** of the valve assembly **100**.

(34) In some embodiments, the cap **10** includes a base **15**. The base **15** is generally configured to couple the cap **10** to the body **16a** of the valve housing **16**. In the current example, the base **15** is

defined by a distal end of the second portion of the sidewall **14b**. The base **15** is configured to couple to the distal end of the first portion of the sidewall **14a**. In some other embodiments, where the cap **10** does not include a portion of the sidewall, the base can be defined by an outer perimetric boundary of the cap **10**. The base can be configured to form various types of connections with the valve body **16a**. For example, the base **15** can be configured to form a snap fit **9** with the valve body **16a** (as illustrated in FIGS. **3-4**). In alternative embodiments, the base **15** can be threadably engageable with the valve body **16a**. The base **15** can be adhered to the valve body **16a** using an adhesive or can be welded to the valve body **16a**. Other types of connections are also contemplated such as a bayonet connection, for example. In some alternate embodiments, the cap **10** can form a unitary component with the valve body **16a** and thus would lack a base (**15**).

(35) As illustrated, the base **15** extends laterally outward from the central axis X (illustrated in FIG. **1**). Further, the first sidewall portion **14a** of the valve body **16a** is positioned laterally outward from the second sidewall portion **14b** of the cap **10**. In alternative embodiments, the base **15** can extend laterally towards the central axis X such that the first sidewall portion **14a** of the body **16a** is positioned laterally inward relative to the second sidewall portion **14b**. In alternative embodiments, the base **15** can define a lateral surface of the cap **10**, which faces towards the first axial end **1**. In some embodiments the first sidewall portion **14a** and the second sidewall portion **14b** are not offset in the lateral direction.

(36) The cap **10** generally has the piercing protrusion **12** (**12a-c** as illustrated in FIGS. **3-6**) extending from the inner lateral surface **11** of the cap **10** towards the first axial end **1**. The piercing protrusion(s) **12** are generally configured to pierce the film **20**. The piercing protrusion(s) **12** can taper towards the first axial end **1**. In alternative embodiments, the piercing protrusion(s) can remain at the same width or diameter for at least a portion of the length or for the entire length of the piercing protrusion(s).

(37) The valve assembly **100** is generally configured to allow gases and/or liquids from inside the enclosure E to escape to the outside environment when the environment inside the enclosure E undergoes a relative pressure spike. Upon a pressure event inside the enclosure E that reaches a burst threshold pressure, the film **20** is configured to burst open.

(38) The piercing protrusion(s) **12** can be configured to pierce the film **20** when a pressure differential between the enclosure opening **19** and the environmental opening **18** rises above the burst threshold. The piercing protrusion(s) **12** can have a piercing tip **13** (**13a-b** illustrated in FIG. **6**). The piercing tip(s) **13** may be relatively sharp to pierce the film **20**. In alternative embodiments, the piercing tip(s) **13** may be blunted to effectuate tears in the film **20**, for example, to increase the area of the film **20** which bursts upon piercing.

(39) In the current example, under normal pressure conditions, the film **20** is generally, although not necessarily perfectly, planar. Upon a pressure spike within the enclosure E above a burst threshold, the pressure expands the film **20** towards the piercing protrusion(s) **12** until the piercing protrusion(s) **12** punctures the film **20**. This allows airflow to pass through the film **20** relatively quickly and exit the valve housing **16** through the environmental opening(s) **18**. Other valve assembly configurations are contemplated, however.

(40) In some embodiments, the tip(s) **13** of the piercing protrusion(s) **12** can extend beyond the base **15** towards the first axial end **1**. The tip(s) **13** can extend beyond the base **15** by at least 1 mm. In alternative embodiments, the tip(s) can extend beyond the base **15** by at least 2 mm, at least 3 mm, at least 4 mm, at least 5 mm, at least 6 mm, at least 7 mm, at least 8 mm, at least 9 mm, or at least 10 mm. In further alternative embodiments, the tip(s) **13** can extend beyond the base **15** by less than less than 10 mm, less than 9 mm, less than 8 mm, less than 7 mm, less than 6 mm, less than 5 mm, less than 4 mm, less than 3 mm, less than 2 mm, or less than 1 mm. In further alternative embodiments, the tip(s) do not extend beyond the base **15**.

(41) The valve assembly **100** generally defines an axial distance L between the film **20** and the inner lateral surface **11** of the cap **10** when the film is in a generally planar configuration and/or in

an undeflected state. The minimum axial distance L can be configured to accommodate successful bursting of the film **20** at the burst threshold. The minimum axial distance L can be configured to accommodate consistent, predictable bursting of the film **20** across a plurality of valve assemblies **100** at the burst threshold. The axial distance L is generally greater than 4.4 mm. In one embodiment, the axial distance L can be greater than or equal to 6.4 mm. In alternative embodiments, the axial distance L can be at least 5 mm, at least 7 mm, or at least 9 mm. The maximum axial distance is not generally limited except by the space that is configured to accommodate the installed valve assembly **100**. In embodiments, the axial distance L can be less than 30 mm, less than 27 mm, less than 25 mm, less than 22 mm, less than 20 mm, less than 17 mm, or less than 15 mm.

(42) In the current example a cap height, Z (illustrated in FIG. 4) is defined in the axial direction from the inner lateral surface **11** of the cap **10** across the base **15**. The cap height Z generally excludes the length of the piercing protrusion **12** extending beyond the base **15**, if any. The cap height Z can be at least 2 mm. In alternative embodiments, the cap height Z can be at least 2 mm, at least 3 mm, at least 4 mm, at least 5 mm, at least 6 mm, at least 7 mm, at least 8 mm, at least 9 mm, at least 10 mm, at least 11 mm, at least 12 mm, at least 13 mm, at least 14 mm, or at least 15 mm. In further alternative embodiments, the cap height Z can be less than 15 mm, less than 14 mm, less than 13 mm, less than 12 mm, less than 11 mm, less than 10 mm, less than 9 mm, less than 8 mm, less than 7 mm, less than 6 mm, less than 5 mm, less than 4 mm, less than 3 mm, or less than 2 mm. However, it will be appreciated that the cap height can be zero in embodiments where the cap does not incorporate a portion of the housing sidewall **14**.

(43) A ratio R1 of the axial distance L to the expansion diameter **21** can be selected to increase the likelihood of a successful burst during a high-pressure event. While not wishing to be bound by theory, it is believed that the greater the expansion diameter **21** of the film **20**, the greater the axial distance L required to ensure bursting of the film. This is described in more detail with respect to FIGS. 6 and 7, below. The ratio of the axial distance L to the expansion diameter **21** will generally be greater than 0.14. In some embodiments, the ratio of the axial distance L to the expansion diameter **21** is greater than or equal to 0.20, 0.23, or 0.25. The maximum ratio of the axial distance L to the expansion diameter **21** is not particularly limited except by the space that is configured to accommodate the installed valve assembly **100**. In embodiments, the ratio of the axial distance L to the expansion diameter **21** can be less than or equal to 1.0, less than or equal to 0.90, less than or equal to 0.80, less than or equal to 0.70, less than or equal to 0.50.

(44) In some embodiments, a ratio R2 of the axial distance L to the expansion area **20a** can be selected to increase the likelihood of a successful burst during a high-pressure event. In embodiments, the ratio of the axial distance L to the expansion area can be greater than 0.005, or greater than or equal to 0.007, 0.008 or 0.009, where the expansion area **20a** is measured in square mm and the axial distance is measured in mm. The ratio of the axial distance L to the expansion area **20a** is not particularly limiting, but can be less than or equal to 0.04 in some embodiments. In embodiments, the ratio of the axial distance L to the expansion area **20a** can be, for example, less than or equal to 0.03, or 0.02.

(45) The burst threshold can be at least 50 mbar. In embodiments, the burst threshold can be at least 50 mbar, at least 60 mbar, at least 70 mbar, at least 80 mbar, at least 90 mbar, at least 100 mbar, at least 150 mbar, at least 200 mbar, at least 250 mbar, or at least 300 mbar. In further embodiments, the burst threshold can be less than 300 mbar, less than 250 mbar, less than 200 mbar, less than 150 mbar, less than 100 mbar, less than 90 mbar, less than 80 mbar, less than 70 mbar, less than 60 mbar, or less than 50 mbar.

(46) Airflow through the airflow pathway A is generally configured to increase after the film **20** is pierced compared to airflow through the airflow pathway A before the film **20** is pierced. Such increase in airflow may be configured to relieve pressure relatively quickly during a high-pressure event in the enclosure to which the valve assembly is coupled. In embodiments where the film is

breathable, airflow through the airflow pathway A can be configured to increase by at least 12 times after the film **20** is pierced compared to airflow through the airflow pathway A before the film **20** is pierced. Such increase in airflow is calculated at a set pressure differential of 100 mbar and a temperature of 25 degrees Celsius. The pressure differential is measured between the pressure of the ambient environment and the pressure of the enclosure to which the valve assembly is coupled.

(47) For example, a maximum airflow through the valve assembly **100** before the film **20** is pierced may be 6.4 standard liters per minute (SLPM). A minimum airflow through the valve assembly **100** after the film **20** is pierced may be 600 SLPM. Thus, the airflow is increased by at least 94 times. In another example, the maximum airflow through the valve assembly **100** before the film **20** is pierced may be 44 SLPM. The minimum airflow through the valve assembly **100** after the film **20** is pierced may be 600 SLPM. Thus, the airflow is increased by at least 13 times. In alternative embodiments, airflow can be configured to increase by at least 25 times, at least 50 times, at least 60 times, at least 75 times, at least 100 times, at least 150 times, at least 200 times, at least 300 times, or at least 400 times after the film **20** is pierced compared to airflow before the film **20** is pierced. In further alternative embodiments, airflow can be configured to increase by less than 400 times, less than 300 times, less than 200 times, less than 150 times, less than 100 times, less than 75 times, less than 60 times, less than 50 times, or less than 25 times after the film **20** is pierced compared to airflow before the film **20** is pierced. In each of the examples the airflow is measured at a pressure differential of 100 mbar.

(48) Schematics of FIGS. **6** and **7** can help illustrate the difference that the axial distance L between the film and an inner lateral surface of the cap can have on the ability of the film **220/320** to successfully burst (by puncture from a piercing protrusion(s) and a tip(s)). In FIG. **6**, a schematic cap **210** having an inner lateral surface **211** is illustrated. The axial distance L of the valve assembly when the film **220** is generally planar is illustrated. As the pressure differential acting on the film **220** rises, the film **220** flexes or expands from a generally horizontal position to the position as illustrated in FIG. **6**. The inner lateral surface **211** provides structural support to the film **220** such that the piercing protrusion(s) **212** fails to puncture the film **220**. Instead, the film **220** surrounds the piercing protrusion(s) **212** and the tip(s) **213** and settles against the inner lateral surface **211** of the cap **210**. The inner lateral surface **211** opposes the force of the pressure on the film **220**. Thus, the film **220** may not reliably burst at or above the burst threshold.

(49) FIG. **7** illustrates an embodiment is consistent with the technology disclosed herein. The axial distance L is greater than the axial distance depicted above with reference to FIG. **6**. In the current example, the relatively increased axial distance L between the inner lateral surface **311** and the film **320** reduces the opportunity for the inner lateral surface **311** to provide structural support to the film **320** that would oppose the forces on the film resulting from a high-pressure event. Instead, the film **320** encounters the piercing protrusion(s) **312** and the tip(s) **313**, continues to expand or flex during the high-pressure event, and without the support of the inner lateral surface **311** or any other support, the film **320** is successfully punctured and bursts. This allows pressure release during a high-pressure event.

(50) Returning again to FIGS. **1-4**, upon puncturing of the film **20**, the film **20** bursts in a single opening, or bursts in multiple openings created in the film **20**. Airflow through the burst film **20** is generally related to (1) the pressure differential between the housing and the ambient environment and (2) the total flow area defined by the burst opening or openings in the film **20**. The ability of the piercing protrusion(s) **12** to pierce the film **20** is generally related to (1) the expansion diameter of the film and (2) the axial distance L between the inner lateral surface and the film **20**. Some other factors which can influence the ability of the piercing protrusion(s) **12** to pierce the film **20** include (a) a distance Y (see FIG. **4**) between the film **20** and the piercing protrusion(s) **12**, (b) a piercing protrusion length P, (c) a void space **17** (illustrated in FIG. **3**), (d) breathability of the film **20**, (e) other material properties of the film **20**, (f) an angle of the piercing tip(s), and (g) the

location of the piercing tip(s) relative to the expansion area **20a** of the film **20**. Some of these factors are discussed further herein.

(51) The valve assembly **100** may define the distance Y between the film **20** and the piercing protrusion(s) **12**. The distance Y may be configured to allow the film **20** to flex or expand such that the film **20** is not punctured unless there is a high-pressure event in the valve housing **16**. The distance Y may also be configured to allow the film **20** to flex or expand such that the film **20** has an increased surface tension and thus the film **20** is more easily punctured by the piercing protrusion(s) **12**. The distance Y can be 1 mm to 3 mm. In alternative embodiments, the distance Y between the film **20** and the piercing protrusion(s) **12** can be at least 1 mm, at least at least 2 mm, at least 3 mm, at least 4 mm, at least 5 mm, at least 6 mm, at least 7 mm, at least 8 mm, at least 9 mm, or at least 10 mm. In further alternative embodiments, the distance Y between the film **20** and the piercing protrusion(s) **12** can be less than less than 10 mm, less than 9 mm, less than 8 mm, less than 7 mm, less than 6 mm, less than 5 mm, less than 4 mm, less than 3 mm, less than 2 mm, or less than 1 mm. In further alternative embodiments, the distance Y is 0 mm.

(52) The valve assembly may define the piercing protrusion length P between the inner lateral surface **11** of the cap **10**, and the tip(s) **13** of the piercing protrusion(s) **12**. The piercing protrusion length P may be configured to enable relatively consistent and successful bursting of the film **20** at the burst threshold. Such piercing protrusion length can be at least 2 mm. In alternative embodiments, the distance P between the inner lateral surface **11** and the tip(s) **13** can be at least 1 mm, at least at least 2 mm, at least 3 mm, at least 4 mm, at least 5 mm, at least 6 mm, at least 7 mm, at least 8 mm, at least 9 mm, at least 10 mm, or at least 15 mm. In further alternative embodiments, the distance P between the inner lateral surface **11** and the tip(s) **13** can be less than 15 mm, less than less than 10 mm, less than 9 mm, less than 8 mm, less than 7 mm, less than 6 mm, less than 5 mm, less than 4 mm, less than 3 mm, less than 2 mm, or less than 1 mm.

(53) The valve assembly **100** can define a void space **17**. The void space **17** is generally configured to allow for flexing or expansion of the film **20** when there is a relatively increased pressure applied to the film **20** through the enclosure opening **19**. The void space **17** is also configured to limit forces on the film **20** that oppose the force of the pressure differential on the film **20**. The void space **17** has no components inhabiting its space. The void space **17** is defined among the piercing protrusion(s) **12**, the housing sidewall **14**, the inner lateral surface **11**, and the expansion area **20a** of the film **20**. In some embodiments, the void space **17** is defined among the piercing protrusion(s) **12**, the housing sidewall **14**, the inner lateral surface **11**, and the circular region defined by the expansion diameter **21**. Omitting structures or components in the void space **17** reduces potential forces that may oppose flexing or expansion of the film **20**. The void space is generally configured such that the film **20** is consistently pierced by the piercing protrusion(s) **12** at the burst threshold in a group of vent assemblies.

(54) FIG. 8 depicts a graph demonstrating the relationship between (1) the ratio of the expansion diameter **21** of the film and the axial distance L (between the inner lateral surface of the cap and the film) and (2) airflow through a valve assembly in response to a burst threshold pressure differential. Valve assemblies tested have configurations generally consistent with FIGS. 1-4. The valve assemblies were generally identical except that the axial distance L was varied such that six different axial distances L were tested, which are reflected below in Table 1. The test was performed on 20 valve assemblies (N=20) for each of the six axial distances L tested at room temperature (23° C.).

(55) TABLE-US-00001 TABLE 1 Valve assemblies having the above axial distances and corresponding ratios of (1) the axial distance/expansion diameter and (2) the axial distance/expansion area. Axial Ratio, R1, axial Ratio, R2, axial Distance L distance/expansion distance/expansion (mm) diameter area 4.4 0.14 0.005 6.4 0.20 0.008 7.4 0.23 0.009 8.4 0.26 0.010 9.4 0.29 0.011 12.4 0.38 0.015

(56) For each of the vent assemblies tested, the film is a breathable membrane. The film is circular,

and the expansion diameter of the film of each valve assembly is 32.5 mm as measured across the expansion area **20a** of the film **20** (such that the expansion area **20a** of the film **20** has a total flow area of 829.58 square mm). Each valve assembly **100** was tested at a threshold pressure differential of 70 mbar.

(57) For each group of vent assemblies of the six different axial distances L, the mean degassing airflow of the data set is displayed numerically (in liters/second) and with a crosshair icon. The interquartile range (representing 50% of the total raw data) is represented by the rectangular boxes, and the median degassing airflow is graphically represented with a horizontal line through each box. The upper and lower 25% of the data distribution is represented by the vertical lines extending from the box. The asterisks represent outlier data points. The axial distance L is increased from left to right along the x-axis as illustrated in FIG. **8**, while the expansion diameter is constant, resulting in an increasing axial distance-to-expansion diameter ratio from left to right along the x-axis. The axial distances L considered are 4.4 mm, 6.4 mm, 7.4 mm, 8.4 mm, 9.4 mm, and 12.4 mm as illustrated from left to right.

(58) The initial data set (corresponding to an axial distance L of 4.4 mm and a ratio of 0.14) demonstrates a mean airflow upon exposure to the threshold pressure differential across the film that is below 10 L/sec. The data distribution shows that the airflow approaches zero for at least some samples, suggesting that the film was not successfully punctured for one or more samples and that some samples may have been successfully punctured but had relatively small puncture openings through the film. Each of the remaining data sets (corresponding to axial distances of 6.4 mm-12.4 mm, or ratios of 0.20-0.38) result in airflow above 10 L/sec after the film **20** is successfully punctured. Thus, the ratios above 0.14 result in relatively higher resulting airflow through the valve assembly **100** after the film **20** is punctured in response to a high-pressure event in the valve housing **16** compared to the valve assembly having a ratio of 0.14. Valve assemblies having ratios above 0.20 demonstrate even higher resulting airflow, suggesting more reliable puncturing and/or relatively larger puncture openings. For valve assemblies having ratios above 0.20, the outlying data points of the minimum airflow exceed 7 L/sec. As such, valve assemblies having a ratio of greater than 0.20 may be desirable in some implementations.

(59) Further testing of degassing performance was conducted for assemblies having three axial distances L of 4.4 mm, 12.4 mm, and 7.6 mm, corresponding to R1 ratios of 0.14, 0.38, and 0.23, respectively. Degassing performance was assessed at two different temperatures: 23° C. and 125° C. and the results are reflected in FIG. **9**. Valve assemblies tested had configurations consistent with those described above with respect to FIG. **8**.

(60) As with the collected data described with reference to FIG. **8**, in FIG. **9**, the mean degassing airflow of the data set is displayed numerically (in liters/second) and with a crosshair icon. The interquartile range (representing 50% of the total raw data) is represented by the rectangular boxes, and the median degassing airflow is graphically represented with a horizontal line through each box. The upper and lower 25% of the data distribution is represented by the vertical lines extending from the box. The asterisks represent outlier data points.

(61) The data set for the assemblies having an axial distance L of 4.4 mm and a ratio of 0.14 demonstrates a mean airflow upon exposure to the threshold pressure differential across the film that is generally below 10 L/sec at both tested temperatures. The assemblies having an axial distance of 7.6 mm and 12.4 mm generally had relatively higher resulting airflow through the valve assembly **100** after the film **20** is punctured in response to a high-pressure event in the valve housing **16** compared to the valve assembly having a ratio of 0.14.

(62) FIG. **10** is another example valve assembly **400** consistent with the technology disclosed herein. The valve assembly **400** has components, configurations and functions that are generally consistent with those described above with reference to FIGS. **1-4** unless contrary to the current description or figure. As such, the description of FIGS. **1-4** is incorporated by reference herein with respect to FIG. **10**.

(63) The valve assembly **400** generally has a valve housing **416**. The valve housing **416** has a first axial end **401**, a second axial end **402**, and an airflow pathway **404** extending from the first axial end **401** towards the second axial end **402**. The valve housing **416** defines an environmental opening **418** configured for fluid communication with an outside environment. The valve housing **416** defines an enclosure opening **419** towards the first axial end **401**. The enclosure opening **419** is configured for fluid communication with an interior of an enclosure (not currently depicted; see other examples described above). The valve housing **416** generally has a housing sidewall **414** extending in the axial direction.

(64) In the current example, the valve housing **416** has a mounting structure **438** that has an alternate configuration from that depicted in FIGS. 1-4. The mounting structure **438** includes outer circumferential threads. The outer circumferential threads are configured to be received by inner circumferential threads defined by an enclosure or by an interfacing component that ultimately is coupled to an enclosure. It will be appreciated that the mounting structure **438** can have alternate configurations such as those described above. The mounting structure **438** can alternately define bayonet connectors, a snap fit, fastener openings configured to be aligned with mating fastener openings that are configured to receive a fastener, a laterally extending surface configured to be welded to or otherwise adhered to an enclosure, and the like.

(65) The valve housing **416** also has a cap **410** positioned towards the second axial end **402**. The cap **410** has an inner lateral surface **411**. The housing sidewall **414** extends from the inner lateral surface **411** towards the first axial end **401**. In some embodiments, the cap **410** has a cap sidewall **414b** extending from the inner lateral surface **411** towards the first axial end **401**. In such embodiments, the cap sidewall **414b** defines a cap height.

(66) The cap **410** has a piercing protrusion **412** extending from the inner lateral surface **411** towards the first axial end **401**. Each piercing protrusion **412** has a piercing tip **413**. The piercing protrusion **412** can taper towards the first axial end **401**, in various embodiments. The piercing protrusion **412** can have a length of at least 4 mm in various embodiments. The cap **410** has a cap base **415**. The cap base **415** is configured to couple to the body **416b** of the valve housing **416**. The cap sidewall **414b** extends from the inner lateral surface **411** to the cap base **415**. In various embodiments, the piercing protrusion **412** extends beyond the cap base **415** towards the first axial end **401** by at least 1 mm. The cap **410** can have a cap height of at least 4 mm in various embodiments, where the cap height is the distance from the inner lateral surface **411** across the base **415**.

(67) The valve assembly **400** has a film **420** disposed in the valve housing **416** laterally across the airflow pathway **404**. The film **420** is disposed in the valve housing **416** along the airflow pathway **404** between the enclosure opening **419** and the environmental opening **418**. In some embodiments the film **420** is a breathable membrane. In some other embodiments the film **420** is non-breathable. The film **420** generally has an expansion diameter **420a** defining an expansion area having a surface area of 450 square mm to 1500 square mm. In some embodiments the film **420** expansion diameter is at least 32.5 mm.

(68) The piercing protrusion **412** is generally configured to pierce the film **420** when a pressure differential between the enclosure opening **419** and the environmental opening **418** rises above a burst threshold. The burst threshold can be at least 50 mbar, in some embodiments. A distance between the film **420** and the piercing protrusion **412** can be 1 mm to 3 mm in some embodiments. In some embodiments, airflow through the airflow pathway **404** is configured to increase by at least 12 times after the film **420** is pierced compared to airflow through the airflow pathway **404** before the film **420** is pierced.

(69) There is generally an axial distance **L** defined between the film **420** and the inner lateral surface **411** of the cap **410**. In various embodiments the axial distance **L** is greater than 4.4 mm. In some embodiments the axial distance **L** is at least 7.4 mm. In various embodiments, a ratio of the axial distance **L** to the film expansion diameter **420a** is greater than 0.14. In various embodiments, the ratio of the axial distance **L** to the expansion diameter **420a** is at least 0.20. In various

embodiments, a void space **417** is defined among the piercing protrusion **412**, the housing sidewall **414** the inner lateral surface **411**, and the film **420**.

Exemplary Aspects

(70) Aspect 1. A valve assembly comprising: a valve housing having: a first axial end, a cap, a second axial end, and an airflow pathway extending from the first axial end towards the second axial end, wherein the cap is positioned towards the second axial end, an environmental opening configured for fluid communication with an outside environment, and an enclosure opening towards the first axial end configured for fluid communication with an interior of an enclosure; and a film disposed in the valve housing laterally across the airflow pathway and along the airflow pathway between the enclosure opening and the environmental opening, wherein the cap comprises an inner lateral surface and a piercing protrusion extending from the inner lateral surface of the cap towards the first axial end, and wherein the piercing protrusion comprises a piercing tip, and wherein an axial distance between the film and the inner lateral surface of the cap is greater than 4.4 mm.

(71) Aspect 2. The valve assembly of any one of Aspects 1 and 3-16, wherein the axial distance between the film and the inner lateral surface of the cap is at least 7.4 mm.

(72) Aspect 3. The valve assembly of any one of Aspects 1-2 and 4-16, wherein the piercing protrusion is configured to pierce the film when a pressure differential between the enclosure opening and the environmental opening rises above a burst threshold.

(73) Aspect 4. The valve assembly of any one of Aspects 1-3 and 5-16, wherein the burst threshold is at least 50 mbar.

(74) Aspect 5. The valve assembly of any one of Aspects 1-4 and 6-16, wherein a distance between the film and the piercing protrusion is 1 mm to 3 mm.

(75) Aspect 6. The valve assembly of any one of Aspects 1-5 and 7-16, wherein a length of the piercing protrusion is at least 2 mm.

(76) Aspect 7. The valve assembly of any one of Aspects 1-6 and 8-16, wherein the cap comprises a cap sidewall extending from the inner lateral surface towards the first axial end, and wherein the cap sidewall defines a cap height.

(77) Aspect 8. The valve assembly of any one of Aspects 1-7 and 9-16, wherein the cap comprises a cap base, the cap sidewall extends from the inner lateral surface to the cap base, and the piercing protrusion extends beyond the cap base towards the first axial end by at least 1 mm.

(78) Aspect 9. The valve assembly of any one of Aspects 1-8 and 10-16, wherein the cap height is at least 2 mm.

(79) Aspect 10. The valve assembly of any one of Aspects 1-9 and 11-16, wherein the valve housing further comprises a housing sidewall extending between the inner lateral surface and the first axial end.

(80) Aspect 11. The valve assembly of any one of Aspects 1-10 and 12-16, further comprising a void space defined among the piercing protrusion, the housing sidewall, the inner lateral surface, and the film.

(81) Aspect 12. The valve assembly of any one of Aspects 1-11 and 13-16, wherein the film comprises a breathable membrane.

(82) Aspect 13. The valve assembly of any one of Aspects 1-12 and 14-16, wherein airflow through the airflow pathway is configured to increase by at least 12 times after the film is pierced compared to airflow through the airflow pathway before the film is pierced.

(83) Aspect 14. The valve assembly of any one of Aspects 1-13 and 15-16, wherein the film comprises a non-breathable membrane.

(84) Aspect 15. The valve assembly of any one of Aspects 1-14 and 16, wherein a surface area of an expansion area of the film is 450 square mm to 1500 square mm.

(85) Aspect 16. The valve assembly of any one of Aspects 1-15, wherein the piercing protrusion tapers towards the first axial end.

(86) Aspect 17. A valve assembly comprising: a valve housing having: a first axial end, a cap, a second axial end, and an airflow pathway extending from the first axial end towards the second axial end, wherein the cap is positioned towards the second axial end, an environmental opening configured for fluid communication with an outside environment, and an enclosure opening towards the first axial end configured for fluid communication with an interior of an enclosure; and a film disposed in the valve housing laterally across the airflow pathway and along the airflow pathway between the enclosure opening and the environmental opening, wherein the film has an expansion diameter, wherein the cap comprises an inner lateral surface and a piercing protrusion extending from the inner lateral surface of the cap towards the first axial end, and wherein the piercing protrusion comprises a piercing tip, wherein an axial distance is defined between the film and the inner lateral surface, and wherein a ratio of the axial distance to the film expansion diameter is greater than 0.14.

(87) Aspect 18. The valve assembly of any one of Aspects 17-34, wherein the axial distance between the film and the inner lateral surface of the cap is at least 7.4 mm.

(88) Aspect 19. The valve assembly of any one of Aspects 17-18 and 20-34, wherein the ratio of the axial distance to the expansion diameter is at least 0.228.

(89) Aspect 20. The valve assembly of any one of Aspects 17-19 and 21-34, wherein the piercing protrusion is configured to pierce the film when a pressure differential between the enclosure opening and the environmental opening rises above a burst threshold.

(90) Aspect 21. The valve assembly of any one of Aspects 17-20 and 22-34, wherein the burst threshold is at least 50 mbar.

(91) Aspect 22. The valve assembly of any one of Aspects 17-21 and 23-34, wherein a distance between the film and the piercing protrusion is 1 mm to 3 mm.

(92) Aspect 23. The valve assembly of any one of Aspects 17-22 and 24-34, wherein a length of the piercing protrusion is at least 2 mm.

(93) Aspect 24. The valve assembly of any one of Aspects 17-23 and 25-34, wherein the cap comprises a cap sidewall extending from the inner lateral surface towards the first axial end, and wherein the cap sidewall defines a cap height.

(94) Aspect 25. The valve assembly of any one of Aspects 17-24 and 26-34, wherein the cap comprises a cap base, the cap sidewall extends from the inner lateral surface to the cap base, and the piercing protrusion extends beyond the cap base towards the first axial end by at least 1 mm.

(95) Aspect 26. The valve assembly of any one of Aspects 17-25 and 27-34, wherein the cap height is at least 2 mm.

(96) Aspect 27. The valve assembly of any one of Aspects 17-26 and 28-34, wherein the valve housing further comprises a housing sidewall extending between the inner lateral surface and the first axial end.

(97) Aspect 28. The valve assembly of any one of Aspects 17-27 and 29-34, further comprising a void space defined among the piercing protrusion, the housing sidewall, the inner lateral surface, and the film.

(98) Aspect 29. The valve assembly of any one of Aspects 17-28 and 30-34, wherein the film comprises a breathable membrane.

(99) Aspect 30. The valve assembly of any one of Aspects 17-29 and 31-34, wherein airflow through the airflow pathway is configured to increase by at least 12 times after the film is pierced compared to airflow through the airflow pathway before the film is pierced.

(100) Aspect 31. The valve assembly of any one of Aspects 17-30 and 32-34, wherein the film comprises a non-breathable membrane.

(101) Aspect 32. The valve assembly of any one of Aspects 17-31 and 33-34, wherein the film expansion diameter is at least 32.5 mm.

(102) Aspect 33. The valve assembly of any one of Aspects 17-32 and 34, wherein a surface area of an expansion area of the film is 450 square mm to 1500 square mm.

(103) Aspect 34. The valve assembly of any one of Aspects 17-33, wherein the piercing protrusion tapers towards the first axial end.

Claims

1. A valve assembly comprising: a valve housing having: a first axial end, a cap, a second axial end, and an airflow pathway extending from the first axial end towards the second axial end, wherein the cap is positioned towards the second axial end, an environmental opening configured for fluid communication with an outside environment, and an enclosure opening towards the first axial end configured for fluid communication with an interior of an enclosure; and a film disposed in the valve housing laterally across the airflow pathway and along the airflow pathway between the enclosure opening and the environmental opening, wherein the film comprises a perimetric region that has a fixed position relative to the cap, wherein the cap comprises an inner lateral surface and a piercing protrusion extending from the inner lateral surface of the cap towards the first axial end, and wherein the piercing protrusion comprises a piercing tip, and wherein an axial distance between the film and the inner lateral surface of the cap is greater than 4.4 mm.
2. The valve assembly of claim 1, wherein the axial distance between the film and the inner lateral surface of the cap is at least 7.4 mm.
3. The valve assembly of claim 1, wherein the piercing protrusion is configured to pierce the film when a pressure differential between the enclosure opening and the environmental opening rises above a burst threshold, wherein the burst threshold is at least 50 mbar.
4. The valve assembly of claim 1, wherein a distance between the film and the piercing protrusion is 1 mm to 3 mm.
5. The valve assembly of claim 1, wherein a length of the piercing protrusion is at least 2 mm.
6. The valve assembly of claim 1, wherein the cap comprises a cap sidewall extending from the inner lateral surface towards the first axial end, and wherein the cap sidewall defines a cap height that is at least 2 mm.
7. The valve assembly of claim 6, wherein the cap comprises a cap base, the cap sidewall extends from the inner lateral surface to the cap base, and the piercing protrusion extends beyond the cap base towards the first axial end by at least 1 mm.
8. The valve assembly of claim 1, wherein the valve housing further comprises a housing sidewall extending between the inner lateral surface and the first axial end.
9. The valve assembly of claim 8, further comprising a void space defined among the piercing protrusion, the housing sidewall, the inner lateral surface, and the film.
10. The valve assembly of claim 1, wherein a surface area of an expansion area of the film is 450 square mm to 1500 square mm.
11. A valve assembly comprising: a valve housing having: a first axial end, a cap, a second axial end, and an airflow pathway extending from the first axial end towards the second axial end, wherein the cap is positioned towards the second axial end, an environmental opening configured for fluid communication with an outside environment, and an enclosure opening towards the first axial end configured for fluid communication with an interior of an enclosure; and a film disposed in the valve housing laterally across the airflow pathway and along the airflow pathway between the enclosure opening and the environmental opening, wherein the film has an expansion diameter, wherein the film comprises a perimetric region that has a fixed position relative to the cap, wherein the cap comprises an inner lateral surface and a piercing protrusion extending from the inner lateral surface of the cap towards the first axial end, and wherein the piercing protrusion comprises a piercing tip, wherein an axial distance is defined between the film and the inner lateral surface, and wherein a ratio of the axial distance to the film expansion diameter is greater than 0.14.
12. The valve assembly of claim 11, wherein the axial distance between the film and the inner lateral surface of the cap is at least 7.4 mm.

13. The valve assembly of claim 11, wherein the ratio of the axial distance to the expansion diameter is at least 0.20.

14. The valve assembly of claim 11, wherein the piercing protrusion is configured to pierce the film when a pressure differential between the enclosure opening and the environmental opening rises above a burst threshold of at least 50 mbar.

15. The valve assembly of claim 11, wherein a distance between the film and the piercing protrusion is 1 mm to 3 mm.

16. The valve assembly of claim 11, wherein a length of the piercing protrusion is at least 2 mm.

17. The valve assembly of claim 11, wherein the cap comprises a cap sidewall extending from the inner lateral surface towards the first axial end, and wherein the cap sidewall defines a cap height of at least 2 mm.

18. The valve assembly of claim 11, wherein the valve housing further comprises a housing sidewall extending between the inner lateral surface and the first axial end, and a void space defined among the piercing protrusion, the housing sidewall, the inner lateral surface, and the film.

19. The valve assembly of claim 11, wherein the film expansion diameter is at least 32.5 mm.

20. The valve assembly of claim 11, wherein a surface area of an expansion area of the film is 450 square mm to 1500 square mm.
